

# mrr\_project\_bio

2025-11-10

## Classification of Cancer outcome using Genetic and Clinical data

### Introduction

This project investigates the effect of genetic and clinical variables on the survival outcome of breast cancer patients. The data contains records of more than 1000 breast cancer patients from several research institutions. Clinical data contains patient-related and tumor-related information. Additionally mRNA gene expression data is available for each patient. The gene expression data has been processed to include only the top 5000 most variable genes on the transformed scale ( $\log_2(\text{counts} + 1)$ ).

We consider that the main outcome variable of interest is **vital\_status**, which is defined in clinical data.

### Load data

```
load("mrr_bio.Rdata")
```

#### Genes data

```
dim(GeneX)
```

```
## [1] 1231 5000
```

```
str(GeneX)
```

```
## num [1:1231, 1:5000] 0 0 10.92 3.17 5 ...
## - attr(*, "dimnames")=List of 2
## ..$ : chr [1:1231] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157" "A8-A08H-01A-21R-A00Z"
## ..$ : chr [1:5000] "CLEC3A" "SCGB2A2" "CPB1" "GSTM1" ...
```

```
colnames(GeneX)[1:10]
```

```
## [1] "CLEC3A" "SCGB2A2" "CPB1" "GSTM1" "TFF1" "SCGB1D2"
## [7] "KCNJ3" "MUCL1" "LINC00993" "ANKRD30A"
```

```
rownames(GeneX)[1:10]
```

```
## [1] "EW-A2FS-01A-11R-A17B" "OL-A6VR-01A-32R-A33J" "E9-A226-01A-21R-A157"
## [4] "A8-A08H-01A-21R-A00Z" "D8-A27H-01A-11R-A16F" "D8-A3Z6-01A-11R-A239"
## [7] "B6-A1KN-01A-11R-A13Q" "BH-A0DL-01A-11R-A115" "A8-A09X-01A-11R-A00Z"
## [10] "BH-A2L8-01A-11R-A18M"
```

GeneX est un tableau de 1231 lignes par 5000 colonnes :

- Les lignes correspondent aux patients avec le nom de chaque ligne l'identifiant du patient comme EW-A2FS-01A-11R-A17B.
- Les colonnes correspondent aux gènes avec chaque nom de colonne un gène comme CLEC3A, qui sont les 5000 gènes les plus variables d'une personne à l'autre.
- Chaque case correspond à la valeur d'expression (un flottant) d'un certain gène (colonne) pour un certain patient (ligne) -> variables continues

## Clinical data

```
load("mrr_bio.Rdata")
dim(clinical_data)
```

```
## Loading required namespace: S4Vectors
```

```
## NULL
```

```
head(clinical_data[, 1:5])
```

```
## DataFrame with 6 rows and 5 columns
##           initial_weight tissue_type laterality
##           <numeric> <character> <character>
## EW-A2FS-01A-11R-A17B      110      Tumor      Left
## OL-A6VR-01A-32R-A33J      380      Tumor      Right
## E9-A226-01A-21R-A157      510      Tumor      Left
## A8-A08H-01A-21R-A00Z      250      Tumor      Left
## D8-A27H-01A-11R-A16F      170      Tumor      Right
## D8-A3Z6-01A-11R-A239       60      Tumor      Left
##           tissue_or_organ_of_origin age_at_diagnosis
##           <character> <integer>
## EW-A2FS-01A-11R-A17B      Breast, NOS      15302
## OL-A6VR-01A-32R-A33J      Breast, NOS      17702
## E9-A226-01A-21R-A157      Breast, NOS      16511
## A8-A08H-01A-21R-A00Z      Breast, NOS      24137
## D8-A27H-01A-11R-A16F      Breast, NOS      26588
## D8-A3Z6-01A-11R-A239      Breast, NOS      20629
```

```
colnames(clinical_data)
```

```
## [1] "initial_weight"      "tissue_type"
## [3] "laterality"          "tissue_or_organ_of_origin"
## [5] "age_at_diagnosis"    "primary_diagnosis"
## [7] "prior_treatment"     "diagnosis_is_primary_disease"
## [9] "ajcc_pathologic_t"   "morphology"
## [11] "classification_of_tumor" "sites_of_involvement"
## [13] "days_to_last_follow_up" "follow_ups_disease_response"
## [15] "race"                "gender"
## [17] "ethnicity"           "vital_status"
## [19] "age_at_index"        "days_to_birth"
## [21] "age_is_obfuscated"   "bcr_patient_barcode"
## [23] "primary_site"        "disease_type"
```

clinical\_data est un tableau de 1231 lignes par 24 colonnes :

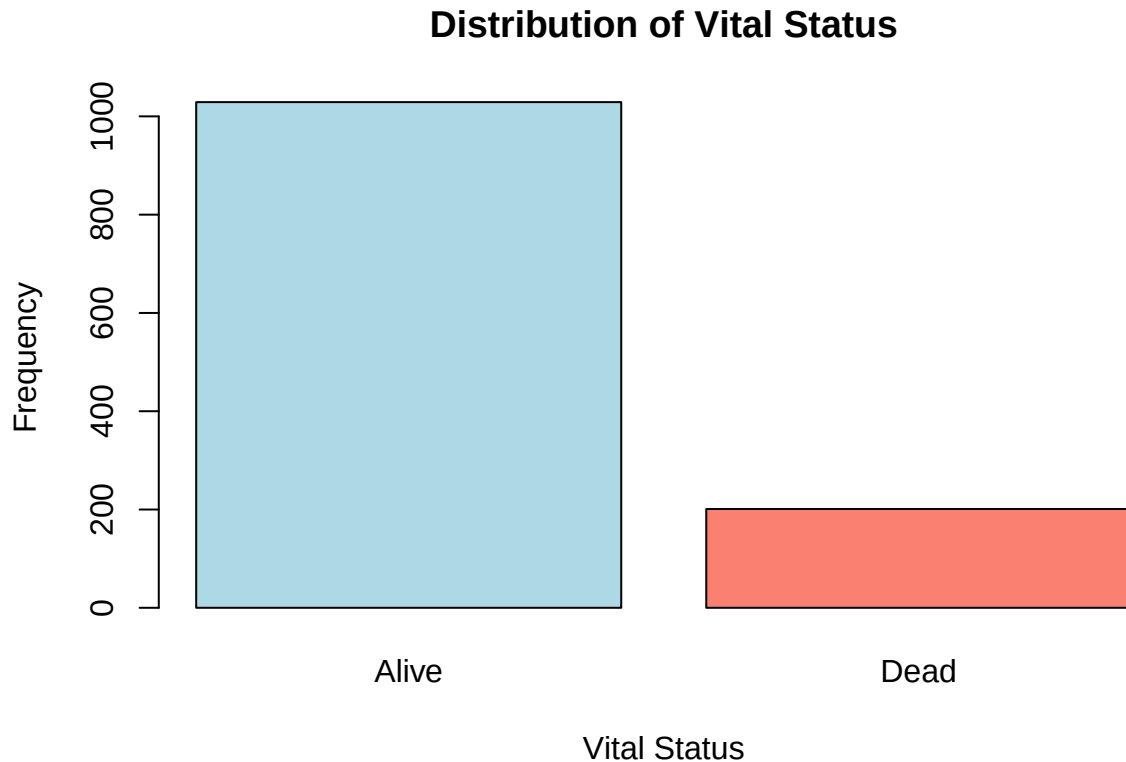
- Les lignes correspondent aux noms des patients comme pour GeneX.
- Les colonnes sont les noms des données cliniques, comme initial\_weight, sites\_of\_involvement et vital\_status.
- Chaque case correspond à la valeur de la donnée clinique pour chaque patient, donc ce peut être un entier ou une chaîne de caractères.

## Outcome variable

The outcome variable is the vital status of the patient (dead or alive) :

```
y <- clinical_data$vital_status
table(y)
```

```
## y
## Alive Dead
## 1029 201
barplot(table(y),
  main = "Distribution of Vital Status",
  xlab = "Vital Status",
  ylab = "Frequency",
  col = c("lightblue", "salmon"))
```



## Exploratory data analysis

### Genes

Pour la tableau de gènes, traçons un premier plot pour chaque gène avec la distribution gaussienne de mort/vivant pour chaque gène. Ensuite nous pouvons prendre la moyenne de ce graphe. Et ensuite tracer toutes les moyennes correspondant à chaque gène dans un graphe commun et vérifier lesquels varient fortement de la médiane (la médiane des moyennes)

### Clinical Data

#### Data cleaning

First model and confusion matrix :

```
x <- clinical_data[, c("initial_weight",
  "classification_of_tumor",
  "morphology")]
y <- clinical_data$vital_status
model_data <- data.frame(x, vital_status = factor(y))
model_data <- na.omit(model_data) # retire les observations incomplètes
model_data$vital_status <- droplevels(model_data$vital_status)
```

First logistic model :

```
model <- glm(vital_status ~ ., data = model_data, family = binomial)
```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

Prediction :

```
pred <- predict(model, type = "response")
positive_class <- levels(model_data$vital_status)[2]
pred_class <- ifelse(pred > 0.5, positive_class, levels(model_data$vital_status)[1])
pred_class <- factor(pred_class, levels = levels(model_data$vital_status))
```

Confusion matrix :

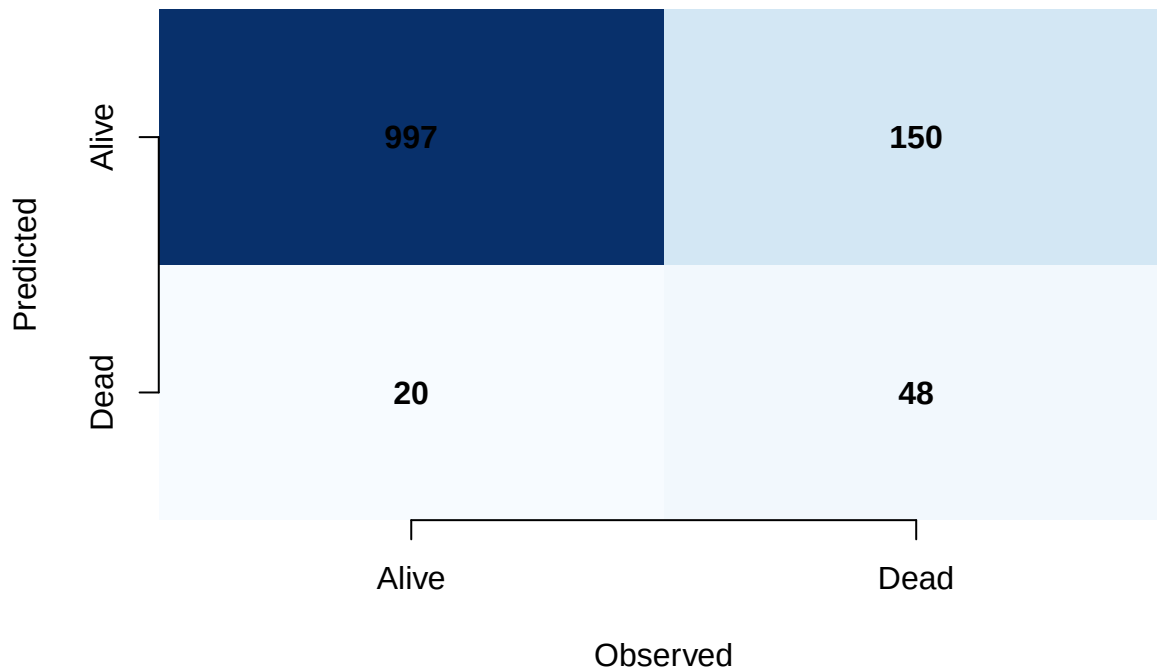
```
conf_mat <- table(Predicted = pred_class, Observed = model_data$vital_status)
conf_mat
```

```
##           Observed
## Predicted Alive Dead
##      Alive   997  150
##      Dead    20   48
```

Visualization of the confusion matrix

```
conf_mat_matrix <- as.matrix(conf_mat)
image(t(conf_mat_matrix[nrow(conf_mat_matrix):1, ]),
      axes = FALSE,
      col = colorRampPalette(c("#f7fbff", "#6baed6", "#08306b"))(64),
      xlab = "Observed",
      ylab = "Predicted",
      main = "Confusion Matrix")
axis(1, at = seq(0, 1, length.out = ncol(conf_mat_matrix)), labels = colnames(conf_mat_matrix))
axis(2, at = seq(0, 1, length.out = nrow(conf_mat_matrix)), labels = rev(rownames(conf_mat_matrix)))
text(rep(seq(0, 1, length.out = ncol(conf_mat_matrix)), each = nrow(conf_mat_matrix)),
     rep(seq(0, 1, length.out = nrow(conf_mat_matrix)), times = ncol(conf_mat_matrix)),
     labels = as.vector(conf_mat_matrix[rev(seq_len(nrow(conf_mat_matrix))), ]),
     col = "black", font = 2)
```

## Confusion Matrix



## Data analysis

### Genes

### Clinical data

**Data cleaning** Voici chaque type correspondant à chaque donnée ainsi que les fréquences de chaque occurrence.

After analysis we can remove redundant data like `age_at_diagnosis`, `days_to_birth` which are like the age in `age_at_index` and `bcr_patient_barcode` which is like the index of the line, and also `disease_type` and `primary_site` because they contain only one one observation which is the same fall all patients.

```
redundant_cols <- c("age_at_diagnosis", "days_to_birth", "bcr_patient_barcode",
                    "disease_type", "primary_site")
clinical_data <- clinical_data[, setdiff(names(clinical_data), redundant_cols)]
cat("Removed columns:", paste(redundant_cols, collapse = ", "), "\n")
```

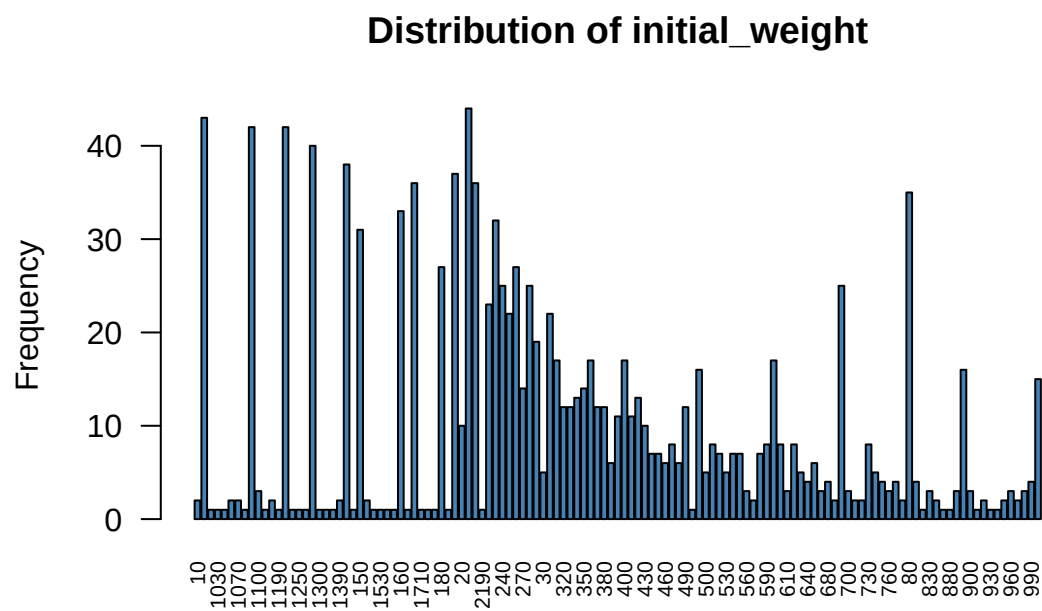
```
## Removed columns: age_at_diagnosis, days_to_birth, bcr_patient_barcode, disease_type, primary_site
```

```
for (col in names(clinical_data)) {
  tbl <- table(as.character(clinical_data[[col]]), useNA = "ifany")
```

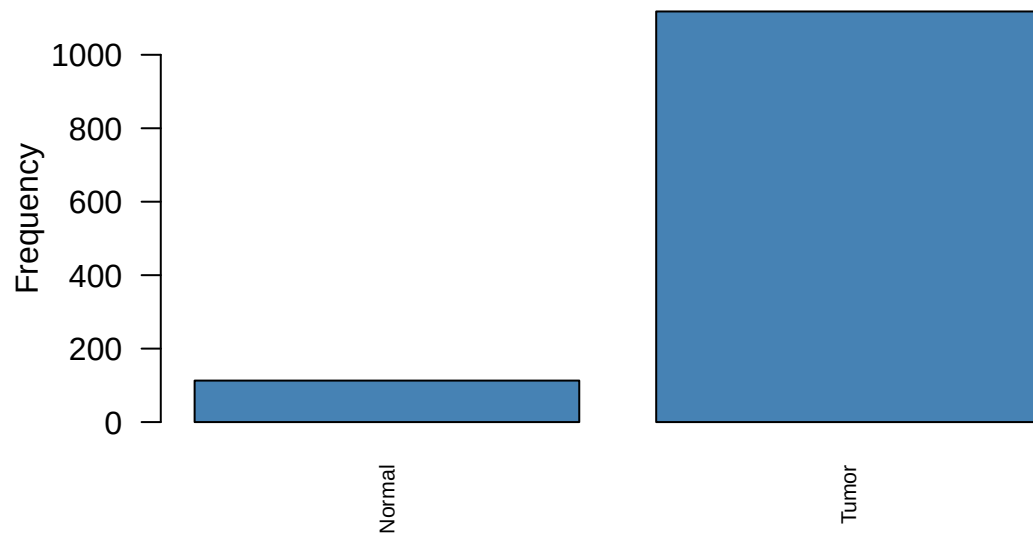
```
  plot.new() # start a fresh plot for each iteration
```

```
  barplot(tbl,
    main = paste("Distribution of", col),
    ylab = "Frequency",
    col = "steelblue",
    las = 2,
    cex.names = 0.7)
```

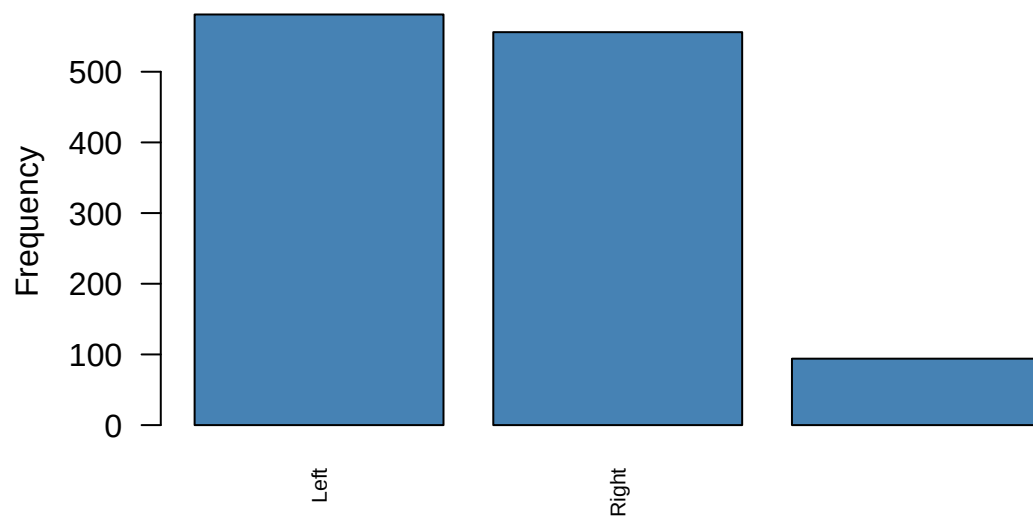
```
}
```



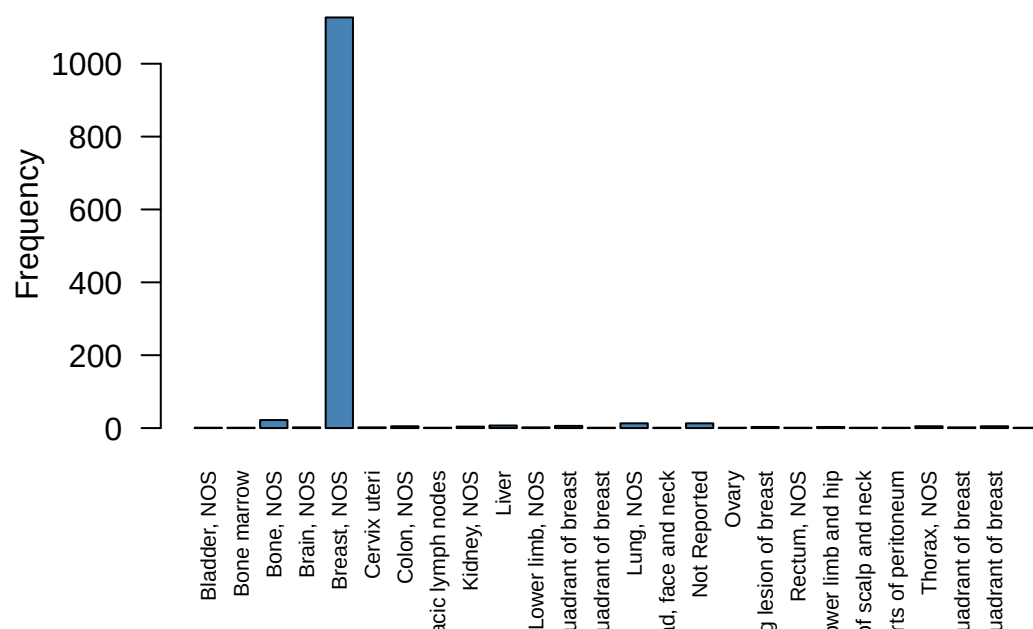
**Distribution of tissue\_type**



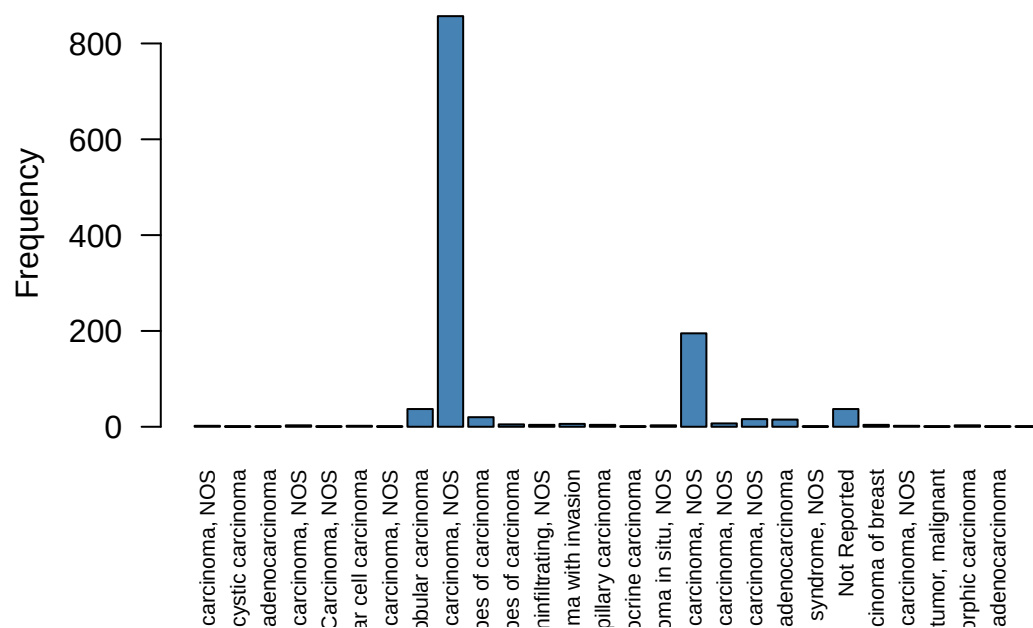
**Distribution of laterality**



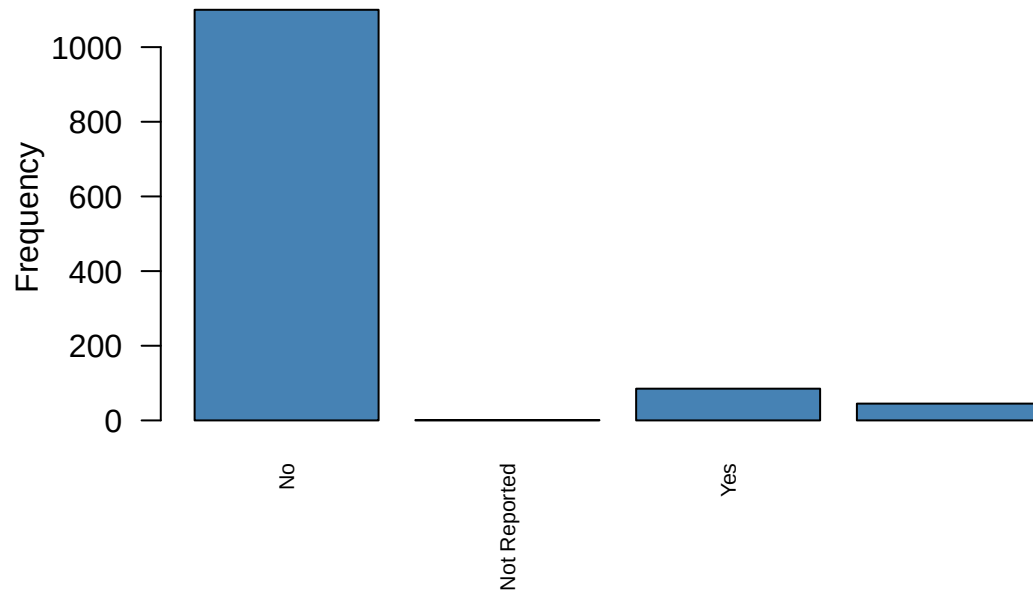
### Distribution of tissue\_or\_organ\_of\_origin



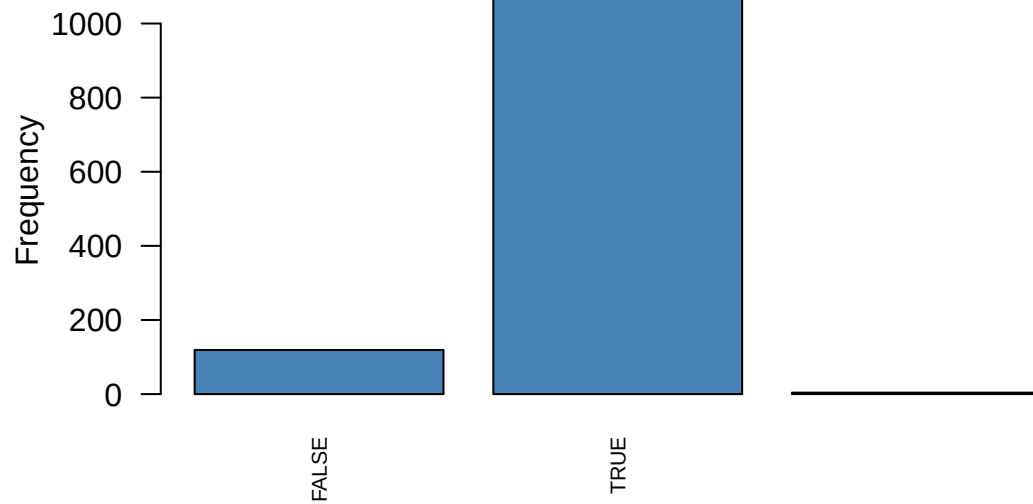
### Distribution of primary\_diagnosis



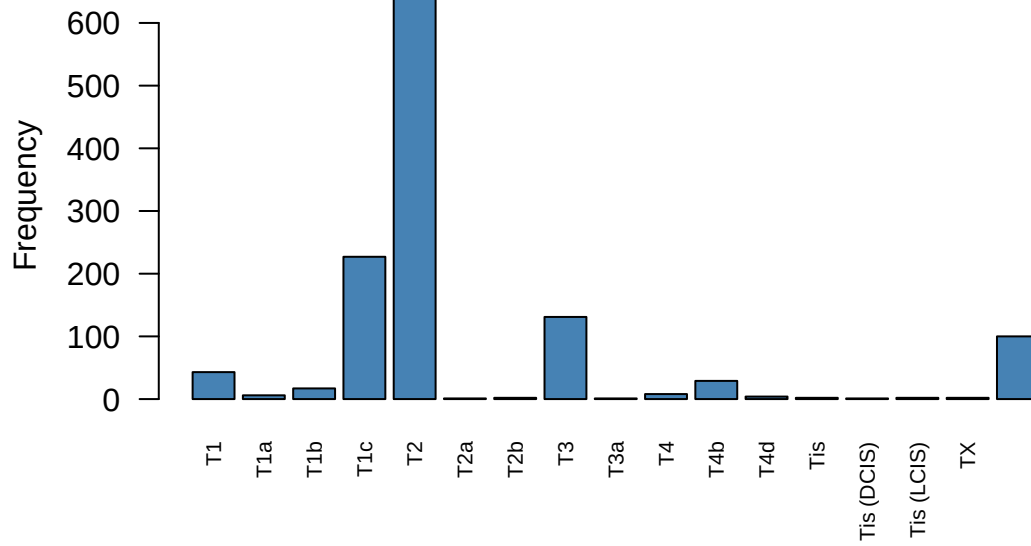
**Distribution of prior\_treatment**



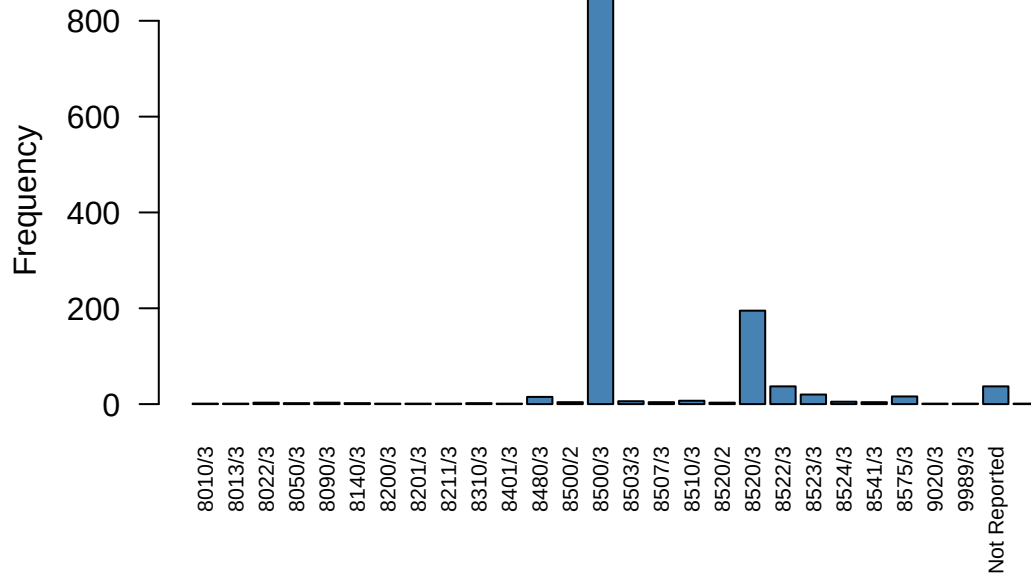
**Distribution of diagnosis\_is\_primary\_disease**



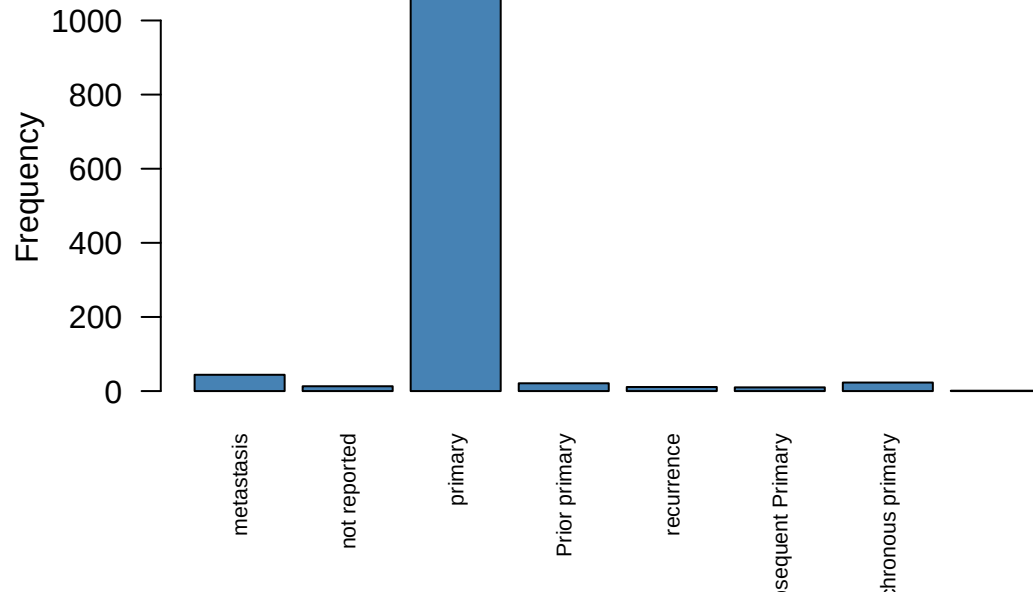
**Distribution of ajcc\_pathologic\_t**



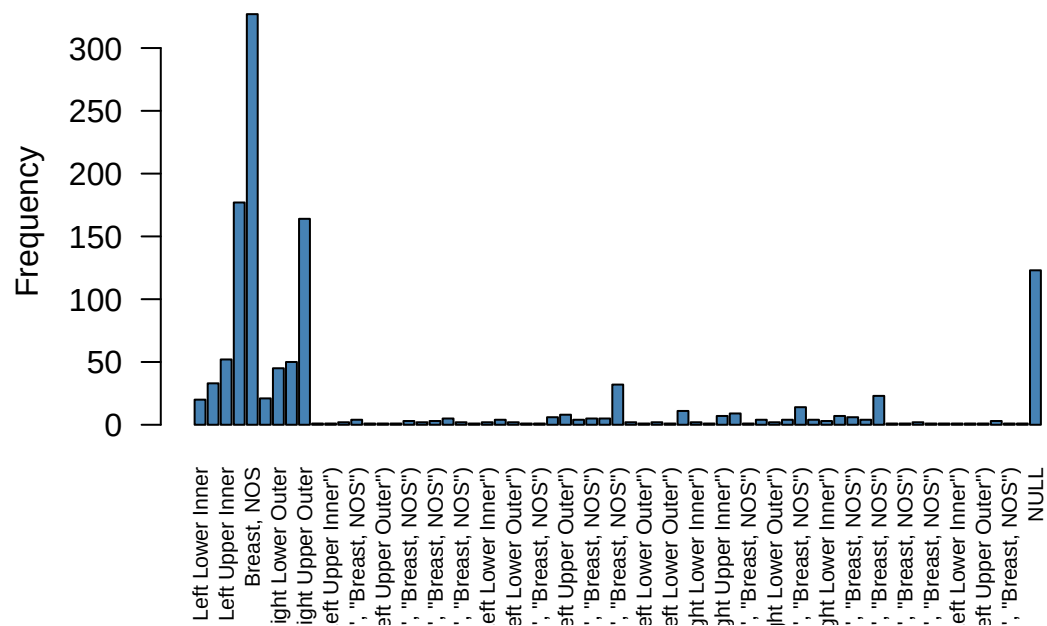
**Distribution of morphology**



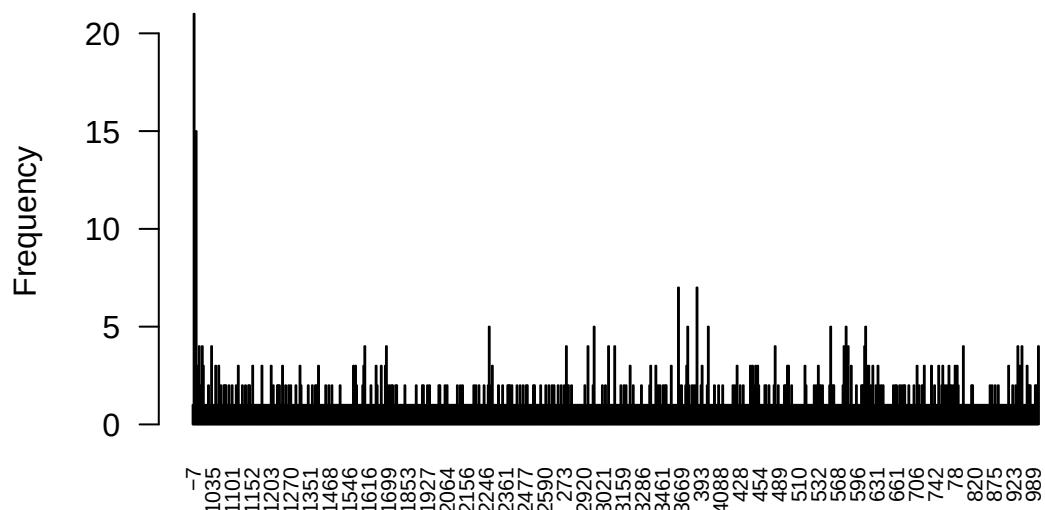
### Distribution of classification\_of\_tumor



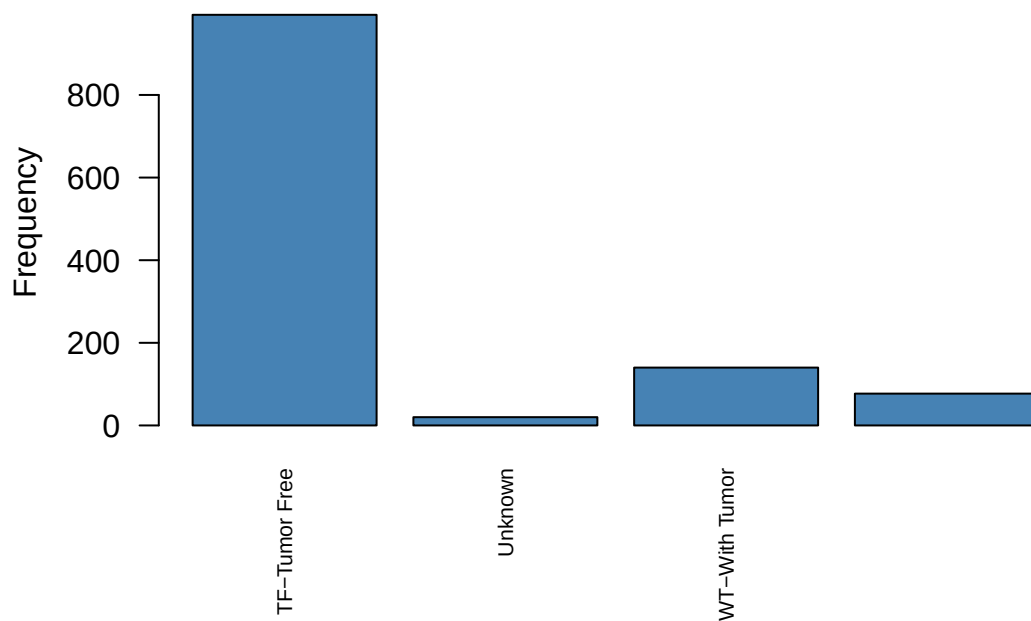
### Distribution of sites\_of\_involvement



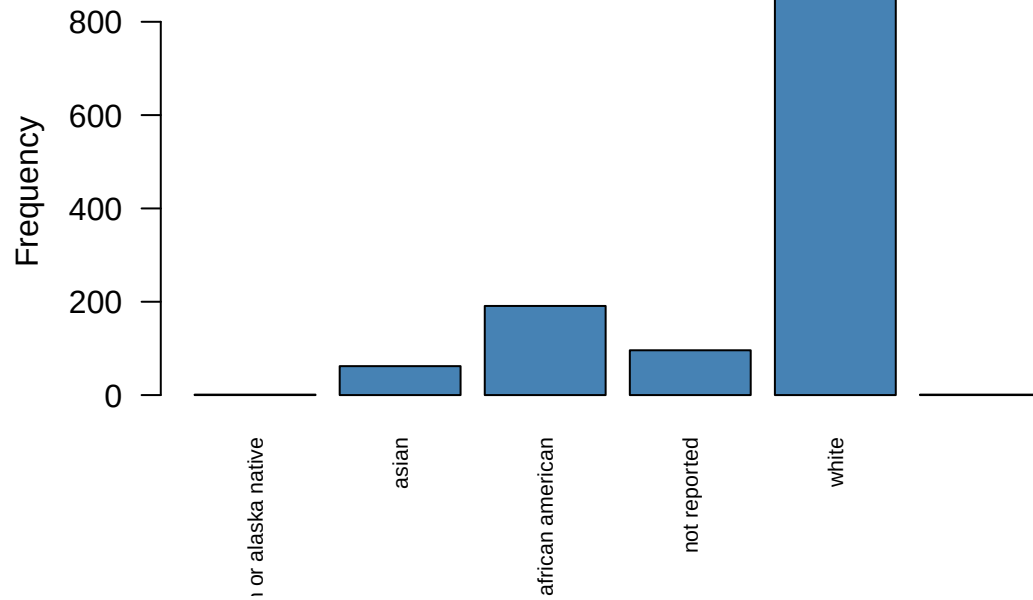
**Distribution of days\_to\_last\_follow\_up**



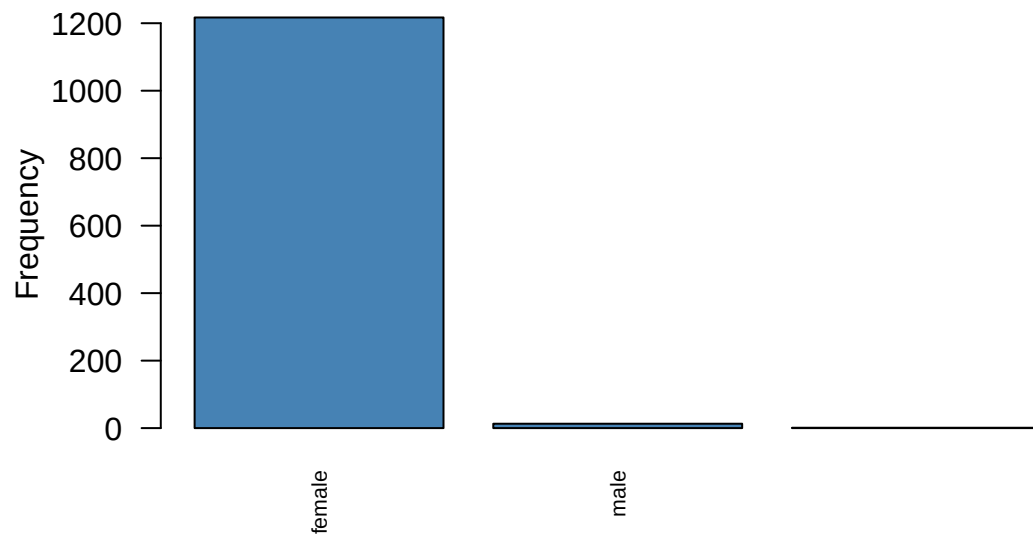
**Distribution of follow\_ups\_disease\_response**

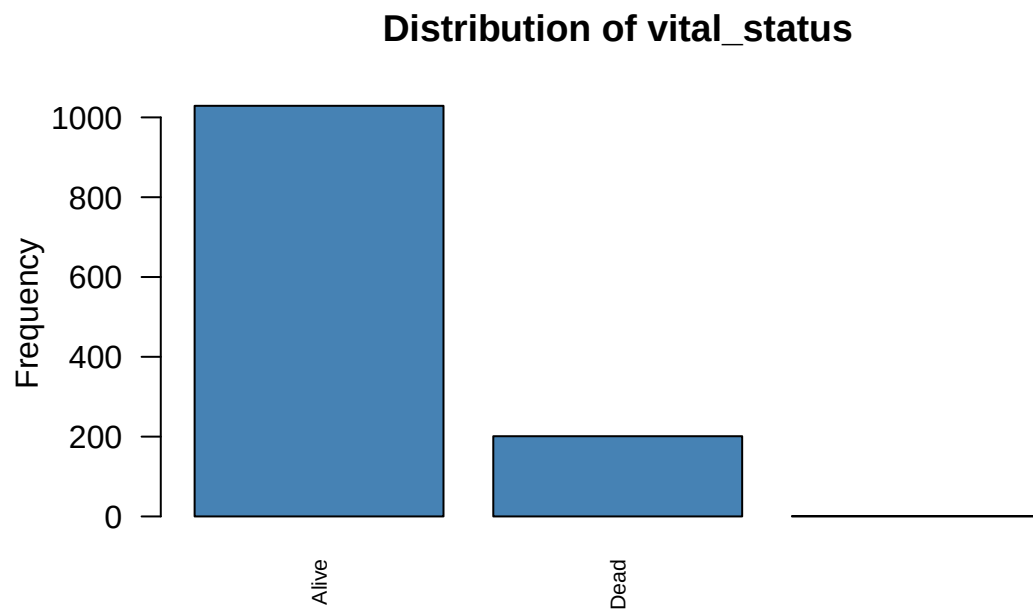
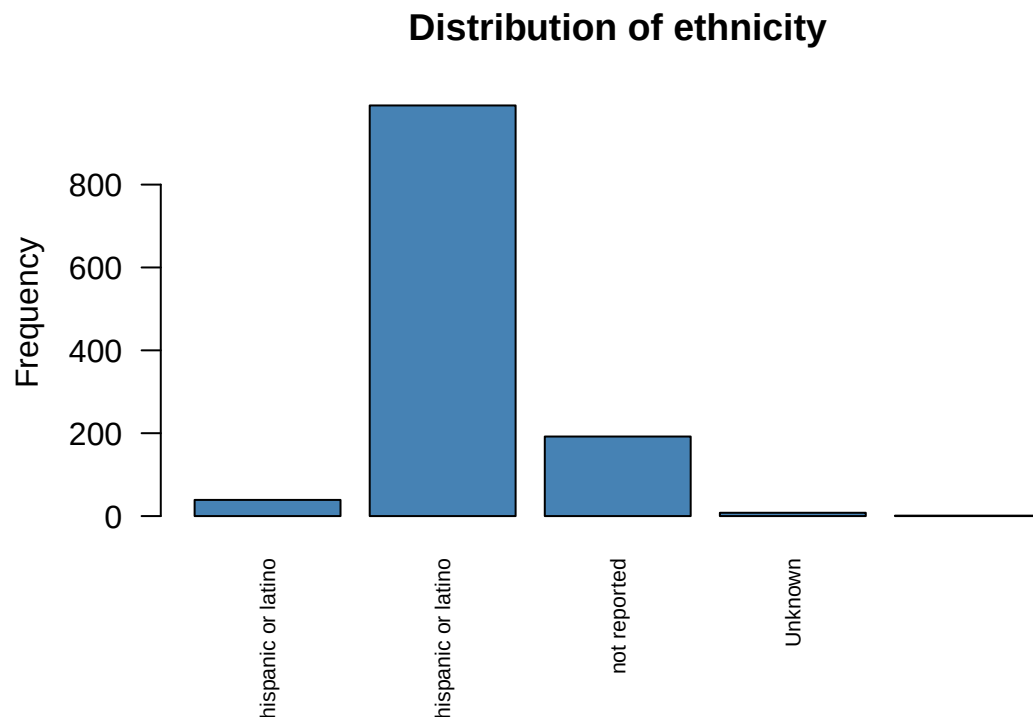


**Distribution of race**

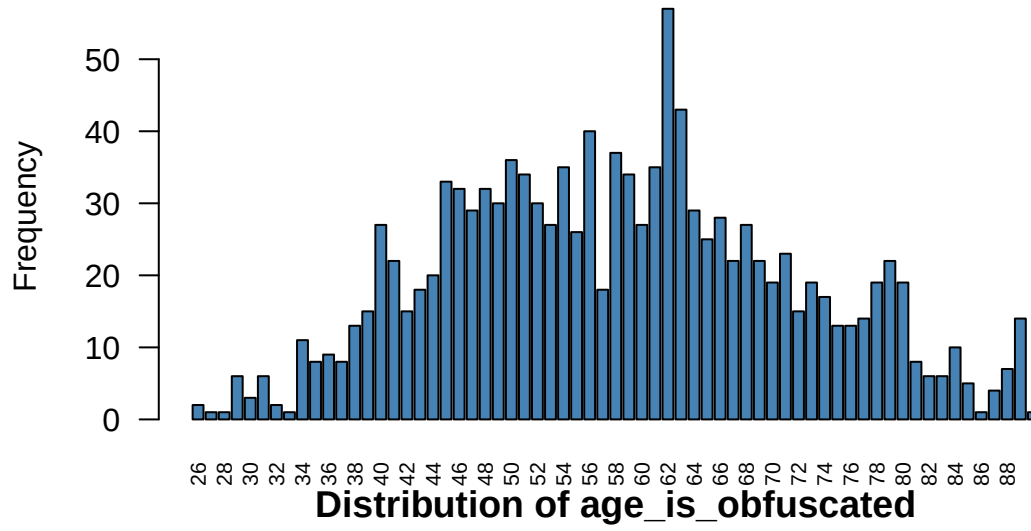


**Distribution of gender**





**Distribution of age\_at\_index**



**Distribution of age\_is\_obfuscated**

